Understanding Analysis, 2nd Edition, Stephen Abbott Chapter 4 - Section 4.4 - Continuous Functions on Compact satisfies

1. (a) Show that $f(x) = x^3$ is continuous on all of $\mathbf{R}$.

   (b) Argue, using Theorem 4.4.5, that f is not uniformly continuous on $\mathbf{R}$.

   (c) Show that f is uniformly continuous on any bounded subset of $\mathbf{R}$.

Solution:

(a) $f(x) = x$ is continuous on all of $\mathbf{R}$. Product of continuous functions is continuous, so $f(x) = x^3$ is continuous as well.

(b) As per the Theorem, we need a sequence (x_n), (y_n) such that $|x_n - y_n| \to 0$ but $|f(x_n) - f(y_n)| \geq \epsilon$ for a choice of ϵ. We'll choose $\epsilon = 1$. Intuitively, as x and y gets closer we place them farther and farther out where x^3 becomes extremely steep.

We'll construct a sequence such that $|x_n - y_n| = \frac{1}{n}$ for all n. We have $|f(x) - f(y)| = |x^3 - y^3| = |x - y||x^2 + xy + y^2|$. Since $|x - y| = \frac{1}{n}$, to make the expression ≥ 1 we need $|x^2 + xy + y^2| \geq n$. Simply choose $x, y \geq n$ for each n, then $|x^2 + xy + y^2| > n^2 > n$, as required. One possible sequence is $(x_n) = n$ and $(y_n) = n + \frac{1}{n}$.

(c) Since the subset is bounded, there exists an $M > 0$ such $|x| \leq M$ for all x. The idea is that the delta for M works for all other $|x| < M$ as well, because x^3 gets steeper farther out from zero and requires smaller and smaller δ.

More formally, do the usual $\epsilon - \delta$ proof to get $|x - y||x^2 + xy + y^2|$. We have $|x|, |y| \leq M$ so $|x - y||x^2 + xy + y^2| \leq 3M^2|x - y|$. Pick $\delta = \frac{\epsilon}{3M^2}$. For an $\epsilon > 0$, this single δ works for all x.

2. (a) Is $f(x) = 1/x$ uniformly continuous on $(0, 1)$?

   (b) Is $g(x) = \sqrt{x^2 + 1}$ uniformly continuous on $(0, 1)$?

   (c) Is $h(x) = x \sin(1/x)$ uniformly continuous on $(0, 1)$?

Solution:

(a) No, as it gets infinitely steeper closer to zero. We'll prove it like problem 1 part (b) and construct (x_n), (y_n) such that $|x_n - y_n| \to 0$ but $|f(x_n) - f(y_n)| \geq 1$.

Construct the sequences such that $|x_n - y_n| = \frac{1}{n}$ for all n. We have $|f(x) - f(y)| = |x - y|\frac{1}{|xy|}$. To make this greater than one, we need $\frac{1}{|xy|} \geq n \implies |xy| \leq \frac{1}{n}$. One such sequence is $(x_n) = \frac{1}{n}$ and $y_n = \frac{2}{n}$, starting from $n = 3$ so that it satisfies the requirement that the domain is $(0, 1)$.

(b) Yes. Do the usual $\epsilon - \delta$ to get

$$\begin{aligned} |\sqrt{x^2+1} - \sqrt{y^2+1}| &= \frac{|x^2 - y^2|}{\sqrt{x^2+1} + \sqrt{y^2+1}} \\ &= |x - y|\frac{|x+y|}{\sqrt{x^2+1} + \sqrt{y^2+1}} \end{aligned}$$

We need to produce a δ so that the above is less than ϵ. Since x and y are bounded by $(0, 1)$, the $\frac{|x+y|}{\sqrt{x^2+1}+\sqrt{y^2+1}}$ term is bounded by some constant c that we don't need to bother computing; the important thing is that it's a constant. Set $\delta = \frac{\epsilon}{c}$, then

$$|x - y|\frac{|x+y|}{\sqrt{x^2+1} + \sqrt{y^2+1}} < \frac{\epsilon}{c} \cdot c = \epsilon$$

This delta works for all y for a choice of ϵ.

Or if we want to compute the actual bound. We have $x, y < 1$ so $|x + y| < 2$. Also, $x, y > 0$ so $\sqrt{x^2+1} + \sqrt{y^2+1} > 2$. Then $\frac{|x+y|}{\sqrt{x^2+1}+\sqrt{y^2+1}} < 1$, so we can set $\delta = \epsilon$. Then

$$|x - y|\frac{|x+y|}{\sqrt{x^2+1} + \sqrt{y^2+1}} < |x - y| < \epsilon$$

(c) Looking at the graph, the answer could possibly be yes. It has a similar problem as $\sin(1/x)$ in that it oscillates wildly closer to zero, but otoh the amplitude gets compressed, so each oscillation covers smaller range of values.

Let's try to prove it's possible and see where it goes. Consider $|x\sin(1/x) - y\sin(1/y)|$. By triangle inequality and the fact that $\sin(1/x) \in [-1, 1]$ for $x \in (0, 1)$, we have

$$|x\sin(1/x) - y\sin(1/y)| \leq |x\sin(1/x)| + |y\sin(1/y)| \leq x + y$$

This bound is only useful if we limit x, y to be small enough. Say, one of $x, y < \epsilon/4$ and $|x - y| < \epsilon/4$, which ensures $x + y < \epsilon/4 + \epsilon/2 < \epsilon$.

What if $x, y \in [\epsilon/4, 1)$? Well, $x \sin(1/x)$ is continuous on $[\epsilon/4, 1]$ (note the closed interval). By Theorem 4.4.7, it's uniformly continuous on $[\epsilon/4, 1]$ and since $[\epsilon/4, 1) \subset [\epsilon/4, 1]$, it's uniformly continuous on $[\epsilon/4, 1)$ as well. Hence there's a single δ_1 that works for this region.

All in all, choose $\delta = \min(\delta_1, \epsilon/4)$.

— Much more elegant proof by chatgpt —
Extend $h(x) = x \sin(1/x)$ to 0 by defining $h(0) = 0$. Now $h(x)$ is continuous on [0, 1], hence uniformly continuous on [0, 1], hence uniformly continuous on (0, 1). This is in fact using the result of Exercise 4.4.13 (Continuous Extension Theorem).

3. Show that $f(x) = 1/x^2$ is uniformly continuous on the set $[1, \infty)$ but not on the set $(0, 1]$.

Solution: First we prove uniform continuity on $[1, \infty)$. $|f(x) - f(y)| = \frac{|x-y||x+y|}{|xy|^2}$. The trick is to handle the $\frac{x+y}{(xy)^2}$ together:

$$\frac{|x-y||x+y|}{|xy|^2} = |x-y|\frac{x+y}{(xy)^2} = |x-y|\left(\frac{1}{xy^2} + \frac{1}{x^2y}\right)$$

Since $x, y \geq 1$, the terms in parentheses is upper-bounded by 2, so we can choose $\delta = \epsilon/2$.

It's not uniformly continuous on $(0, 1]$ for the same reason $1/x$ is not uniformly continuous on $(0, 1]$ (Problem 4.4.2(a)). The exact same argument works, since $1/x^2$ is even steeper.

Let $(x_n) = \frac{1}{n}$ and $(y_n) = \frac{2}{n}$. Pick $\epsilon = 1$. Then $|x_n - y_n| \to 0$ but $|f(x_n) - f(y_n)| = \frac{3n^2}{4} \geq 1$ for $n \geq 2$.

4. True or false.

   (a) If f is continuous on $[a, b]$ with $f(x) > 0$ for all $a \leq x \leq b$, then $1/f$ us bounded on $[a, b]$ (meaning $1/f$ has a bounded range).

   (b) If f is uniformly continuous on a bounded set A, then $f(A)$ is bounded.

   (c) If f is defined on $\mathbf{R}$ and $f(K)$ is compact whenever K is compact, then f is continuous on $\mathbf{R}$.

Solution:

(a) True. By Theorem 4.4.7, f is continuous on $[a, b]$ implies $f([a, b])$ is compact, hence bounded. Since $f([a, b]) > 0$, $1/f$ is defined and bounded on $a \leq x \leq b$.

– Correction by Chatgpt –
I wrote "hence bounded" above, but actually boundedness alone is not enough, we need compactness. If $f[a, b] = (0, 1]$ then it's bounded but $1/f$ is unbounded. But since $f[a, b]$ is compact and $f([a, b]) > 0$, then it cannot have 0 as its limit point.

– Better proof by Chatgpt –
By the Extreme Value Theorem, f attains a minimum m in $[a, b]$. Since $f([a, b]) > 0$, then $m > 0$ so $\frac{1}{f} \leq \frac{1}{m}$.

(b) True. The intuition is that if $f(A)$ were unbounded, since A is bounded then $f(A)$ must contain an infinite slope and that breaks uniform continuity.

Formally, we'll prove the contrapositive: if $f(A)$ is unbounded then f is not uniformly continuous. Choose $x_n \in A$ such that $|f(x_n)| > n$. Since A is bounded, by BWT (x_n) contains a Cauchy subsequence (y_n) such that $f(y_n)$ is also unbounded.

Now suppose f were uniformly continuous. Then for an ϵ there exists a δ that works. Since (y_n) is Cauchy, there exists an N such that $i, j > N \implies |y_i - y_j| < \delta$. But this results in a contradiction. By uniformy continuity $|f(y_i) - f(y_j)| < \epsilon$, which contradicts the fact that $f(y_n)$ is unbounded. Hence f cannot be uniformly continuous.

(c) False. Continuity preserves compactness (Theorem 4.4.1), but preservation of compactness does not imply continuity. Counterexample: let $f(x) = 0$ for $x \neq 1$ and $f(1) = 1$. We have $f(S)$ compact for any set S since $f(S)$ is either $\{0\}$, $\{1\}$, or $\{0, 1\}$. But f is discontinuous at 1.

5. Assume that g is defined on an open interval (a, c) and is uniformly continuous on $(a, b]$ and $[b, c)$ where $a < b < c$. Prove that g is uniformly continuous on (a, c).

Solution: Let $\epsilon > 0$. By uniform continuity on $(a, b]$, there exists $\delta_1 > 0$ such that for all $a < x < y \leq b$, $|x - y| < \delta_1 \implies |f(x) - f(y)| < \frac{\epsilon}{2}$. Similarly, there exists $\delta_2 > 0$ such that for all $b \leq x < y < c$, $|x - y| < \delta_2 \implies |f(x) - f(y)| < \frac{\epsilon}{2}$.

Let $\delta = \min(\delta_1, \delta_2)$. Then even if $x \in (a, b]$ and $y \in [b, c)$, the distances $|a - b|$ and $|b - c|$ are both within δ, so uniform continuity on both intervals ensure the sum is within ϵ. More formally, there are three cases:

1. $a < x < y \leq b$. Then $|x - y| < \delta \leq \delta_1 \implies |f(x) - f(y)| < \frac{\epsilon}{2} < \epsilon$.
2. $b \leq x < y < c$. Then $|x - y| < \delta \leq \delta_2 \implies |f(x) - f(y)| < \frac{\epsilon}{2} < \epsilon$.
3. $a < x < b < y < c$. We have $|x - y| = |x - b| + |b - y|$. Then $|x - b| < \delta \leq \delta_1$ and $|b - y| < \delta \leq \delta_2$ so $|f(x) - f(y)| \leq |f(x) - f(b)| + |f(b) - f(y)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$.

6. Example or impossible with a short explanation.

(a) A continuous function $f : (0, 1) \to \mathbf{R}$ and a Cauchy sequence (x_n) such that $f(x_n)$ is not a Cauchy sequence.

(b) A uniformly continuous function $f : (0, 1) \to \mathbf{R}$ and a Cauchy sequence (x_n) such that $f(x_n)$ is not a Cauchy sequence.

(c) A continuous function $f : [0, \infty) \to \mathbf{R}$ and a Cauchy sequence (x_n) such that $f(x_n)$ is not a Cauchy sequence.

Solution:

(a) Impossible. The issue here is that (x_n) may converge to 0 which is outside the domain, so continuity doesn't help and $f(x_n)$ may be unbounded towards zero. Counterexample: $(x_n) = 1/n$. We have $(x_n) \to 0$ but $f(x_n)$ diverges.

(b) Impossible. We'll prove that $f(x_n)$ must be Cauchy. Given $\epsilon > 0$, we need to produce an N such that for all $n, m > N$, $|f(x_n) - f(x_m)| < \epsilon$.

By uniform continuity, there exists a δ such that $|x_n - x_m| < \delta \implies |f(x_n) - f(x_m)| < \epsilon$.

Since (x_n) is Cauchy, there exists an N such that for all $n, m > N$, $|x_n - x_m| < \delta$.

So the final chain of logic is: (x_n) Cauchy $\implies$ pick an N such that $|x_n - x_m| < \delta \implies |f(x_n) - f(x_m)| < \epsilon$ by uniformy continuity.

(c) True. Let $(x_n) \to L$. We have $L \in [0, \infty)$ and f continuous. By definition of continuity, $(x_n) \to L \implies f(x_n) \to f(L)$ so $f(x_n)$ is Cauchy.

7. Prove that $f(x) = \sqrt{x}$ is uniformly continuous on $[0, \infty)$.

Solution: We have

$$\begin{aligned} |\sqrt{x} - \sqrt{y}| &= \sqrt{|\sqrt{x} - \sqrt{y}|^2} \\ &= \sqrt{x + y - 2\sqrt{xy}} \end{aligned}$$

Where we get rid of the absolute value because a square is non-negative. Now wlog $x \geq y$, then $2\sqrt{xy} \geq 2y$ and $y - 2\sqrt{xy} \leq y - 2y = -y$, so:

$$\sqrt{x + y - 2\sqrt{xy}} \leq \sqrt{x - y}$$

We can pick $\delta = \epsilon^2$.

8. Example or short argument why impossible.

(a) A continuous function defined on [0, 1] with range (0, 1).

(b) A continuous function defined on (0, 1) with range [0, 1].

(c) A continuous function defined on (0, 1] with range (0, 1).

Solution:

(a) Impossible. By Theorem 4.4.1, if a function is continuous on a compact set, then its range must be compact as well.

(b) $|\sin(1/x)|$. It's continuous in $(0, 1)$ because the absolute value function is continuous everywhere, $\sin(1/x)$ is continuous everywhere except 0, and function composition preserves continuity. Alternatively, $\sin^2(1/x)$ or $\cos^2(1/x)$.

(c) **(Wrong solution)** Impossible. If the function is continuous on (0, 1] then it's also continuous on the compact subset [1/2, 1]. By Theorem 4.4.1, its range must be compact on [1/2, 1] as well.

The argument above is wrong because Theorem 4.4.1 only guarantees that $f([1/2, 1])$ is compact but it doesn't say that the range must contain [1/2, 1] as well. For example, it could be that $f([1/2, 1]) = [1/2, 3/4]$.

(Correct solution) TODO.

9. **(Lipschitz Functions).** A function $f : A \to \mathbf{R}$ is called *Lipschitz* if there exists a bound $M > 0$ such that

$$\left| \frac{f(x) - f(y)}{x - y} \right| \leq M$$

for all $x \neq y \in A$. Geometrically speaking, a function f is Lipschitz if there is a uniform bound on the magnitude of the slopes of lines drawn through any two points on the graph of f.

(a) Show that if $f : A \to \mathbf{R}$ is Lipschitz, then it's uniformly continuous on A.

(b) Is the converse statement true? Are all uniformly continuous functions necessarily Lipschitz?

Solution:

(a) Pick $\delta = \frac{\epsilon}{M}$. Then $|f(x) - f(y)| \leq M|x - y| < M\frac{\epsilon}{M} = \epsilon$.

(b) No. Counterexample: $f(x) = \sqrt{x}$ on $[0, \infty)$. f is uniformly continuous (Exercise 7), but as x approaches zero, $\frac{\sqrt{x}}{x}$ approaches infinity.

10. Assume that f and g are uniformly continuous functions defined on a common domain A. Which of the following combinations are necessarily uniformly continuous on A: $f(x) + g(x), f(x)g(x), \frac{f(x)}{g(x)}, f(g(x))$. Assume that the quotient and the composition are properly defined and thus at least continuous.

Solution: $f(x)+g(x)$ is uniformly continuous. Given an $\epsilon > 0$, there exists a δ_f such that $|x-y| < \delta_f \implies |f(x)-f(y)| < \frac{\epsilon}{2}$. Similarly for δ_g. Pick $\delta = \min(\delta_f, \delta_g)$, then $|x-y| < \delta \implies |f(x)-f(y)+g(x)-g(y)| \leq |f(x)-f(y)|+|g(x)-g(y)| < \frac{\epsilon}{2}+\frac{\epsilon}{2} = \epsilon$.

$f(g(x))$ is uniformly continuous. Given an $\epsilon > 0$, there exists a δ_f such that $|x-y| < \delta_f \implies |f(x) - f(y)| < \epsilon$. There also exists a δ_g such that $|x - y| < \delta_g \implies |g(x) - g(y)| < \delta_f$. Pick $\delta = \delta_g$.

$f(x)g(x)$ is not necessarily uniformly continuous. Counterexample: $f(x) = g(x) = x$, then $f(x)g(x) = x^2$ is not uniformly continuous on $\mathbf{R}$.

$\frac{f(x)}{g(x)}$ is not necessarily uniformly continuous. Counterexample: $f(x) = 1$ and $g(x) = x$, then $\frac{f(x)}{g(x)} = \frac{1}{x}$ is not uniformly continuous on $(0, 1)$.